The Nexus Between Excess Returns, Financial Intermediaries and Uncertainty Shocks

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Motivation

- Empirical evidence points at countercyclicality of excess returns
- **New evidence:** magnitude of this dynamics depends on combination of uncertainty shocks AND worsened balance sheet conditions for intermediaries
- Differential responses to an uncertainty shock conditional on the capitalization of the financial sector: worse balance sheet condition $\implies$ larger responses
- **New hypothesis:** intermediaries' precautionary behavior generate differential responses depending on the distance from their financial constraint.

Literature Review

Proposed Mechanism

- Balance sheet of financial intermediaries is significant factor in determining excess returns Literature
- Higher weight on downside risk when balance sheet is deteriorated
- Uncertainty shocks trigger precautionary deleveraging episodes:
  - $\downarrow$ investment in risky productive assets and hence $\downarrow$ output
  - $\uparrow$ of premium on risky asset and $\uparrow$ of excess returns
- Lower capitalization $\implies$ higher risk aversion $\implies$ larger effect.

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Strategy

- Contingency on capitalization of the financial system of response to uncertainty shocks for
  - Excess returns
  - Macro aggregates
  - Intermediaries balance sheet composition
- Local projections (Jordà (2005))
- Capitalization captured by
  $\eta_{t-1} = \%\Delta$ from 5-months MA of $\frac{\text{Common Equity}_{t-1}}{\text{Invested Capital}_{t-1}}$
- Uncertainty indexes: VXO (Bloom (2009)) and U_F^{LMN} (Ludvigson et al. (2021)) Plot
  - Uncertainty shocks identified as in Bloom (2009) for VXO and as in Ludvigson et al. (2021) for U_F^{LMN}. Details

Data

Specification

$$y_{t+h} = \alpha_h + \beta_0^h \epsilon_t^{UNC} + \beta_1^h \epsilon_t^{UNC} \eta_{t-1} + \boldsymbol{\gamma}(L)' \cdot \mathbf{x}_t + u_t$$

$$\widehat{IRF}(h \mid \eta_{t-1} \geq 0) = \widehat{\beta_0^h} + \widehat{\beta_1^h}\overline{\eta}$$
$$\widehat{IRF}(h \mid \eta_{t-1} < 0) = \widehat{\beta_0^h} + \widehat{\beta_1^h}\underline{\eta}$$

- $y_t \in \left\{ (R_i^{exc})_{i=S,B,M}, C, GFCF, \text{Hours worked, ip, Credit } F \text{ to } NF, \text{Treasuries, Net worth, Debt, Mortgages} \right\}$
- $\overline{\eta} = \mathbb{E}[\eta_{t-1} \mid \eta_{t-1} \geq 0]$ and $\underline{\eta} = \mathbb{E}[\eta_{t-1} \mid \eta_{t-1} < 0]$
- $\epsilon_t^{UNC} =$ identified shock to VXO and U_F^{LMN}.

Data

Excess Returns: shock to VXO

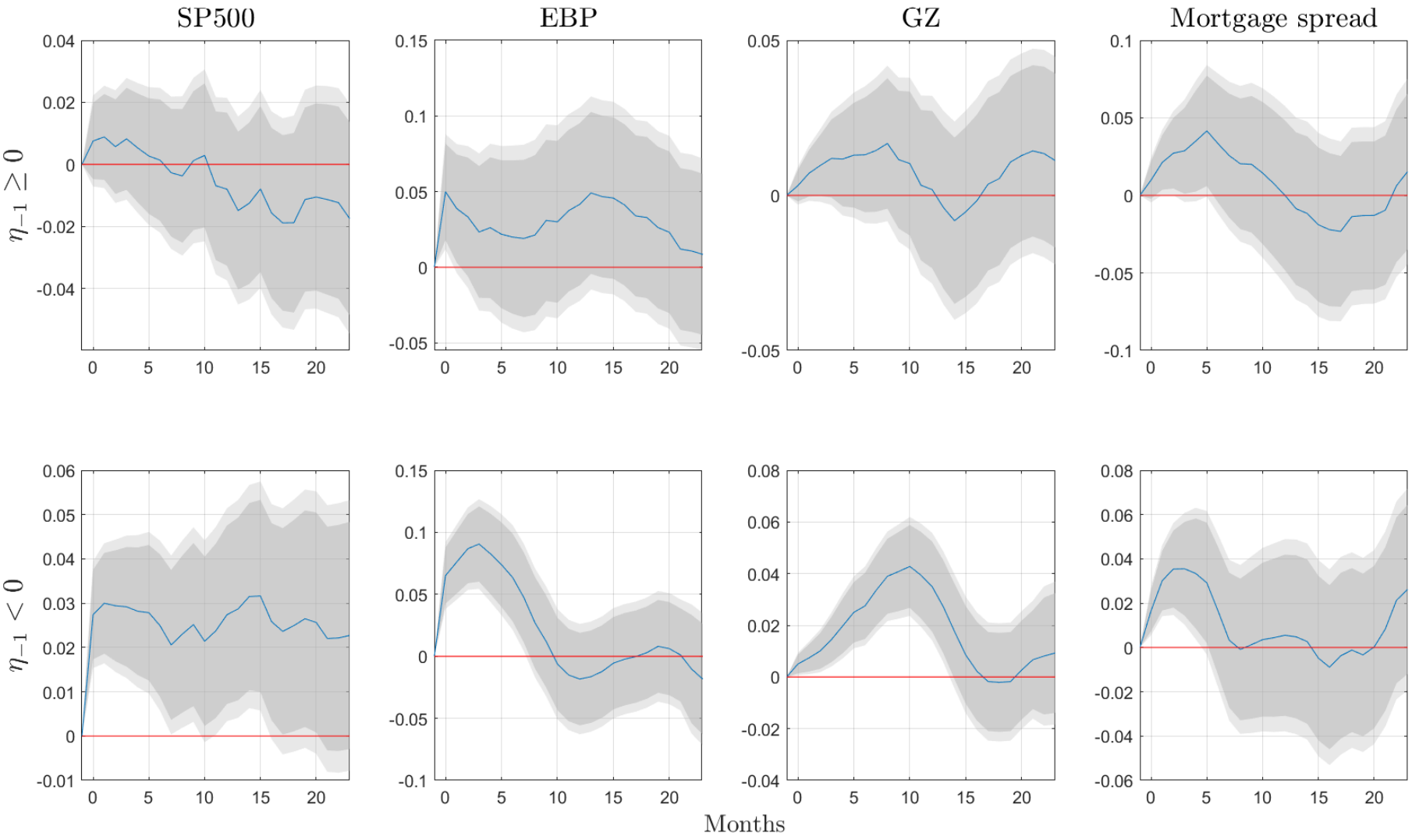

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. LMN

Real Variables: shock to VXO

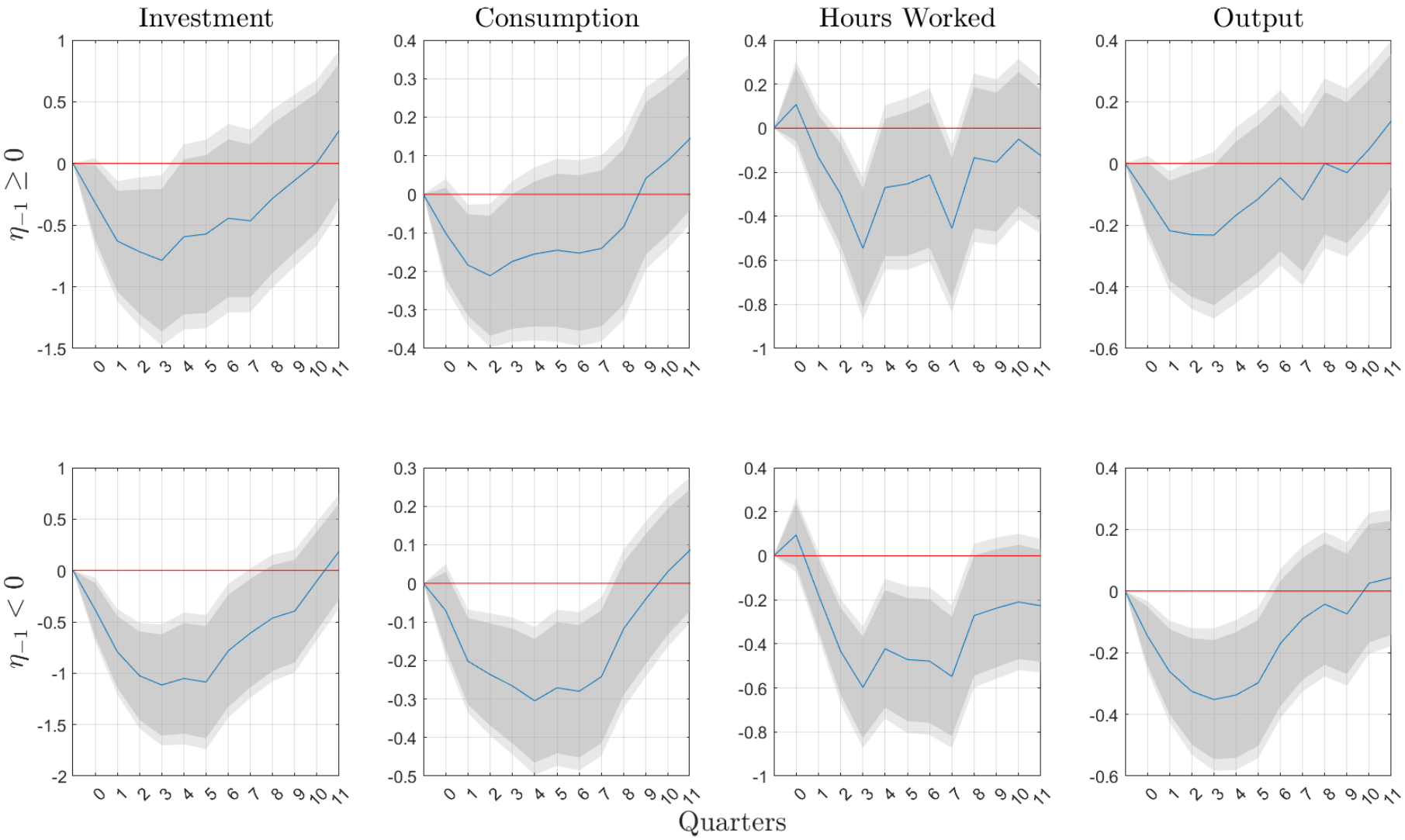

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. LMN

Financial Variables: shock to VXO

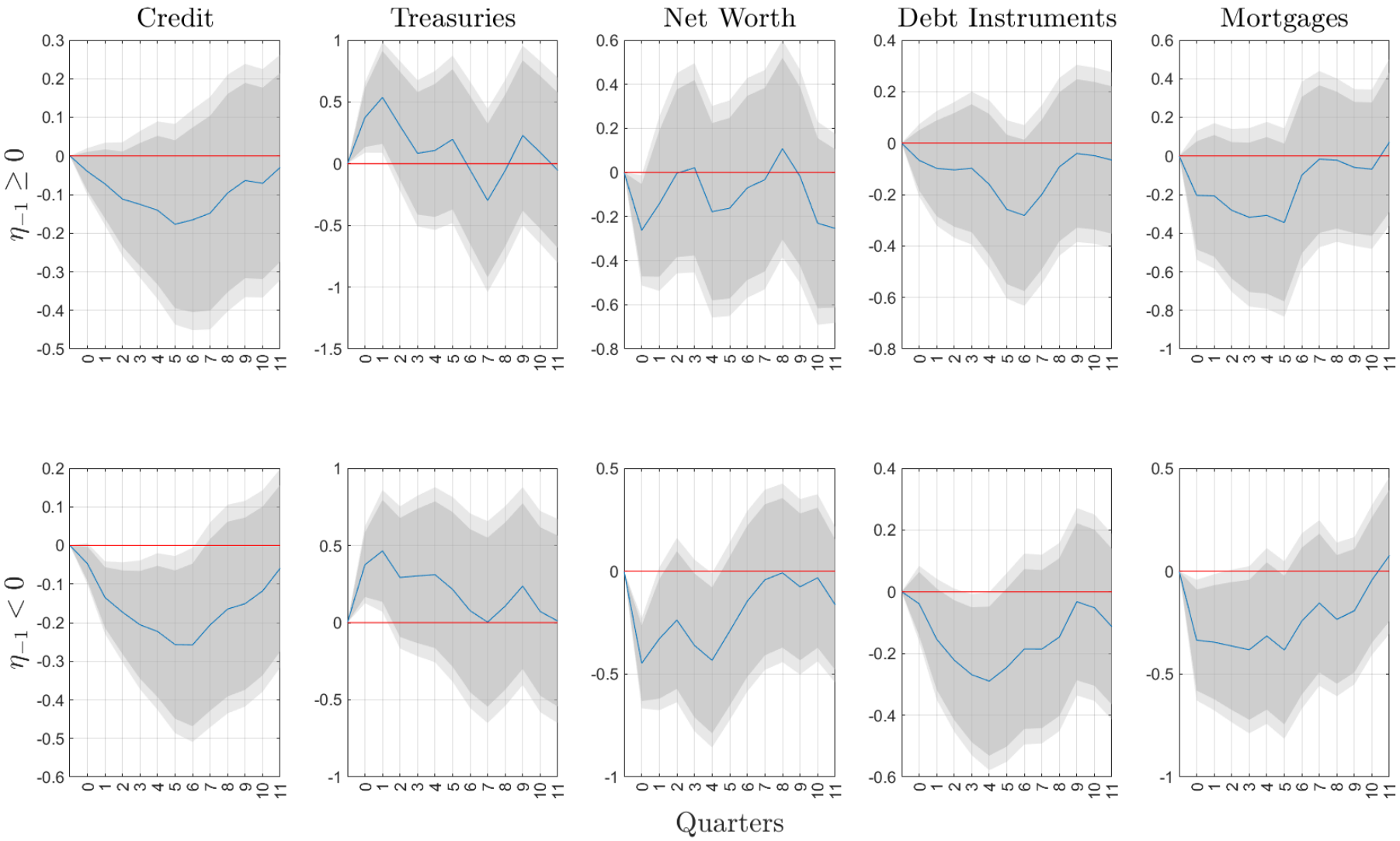

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. LMN Debt Instruments

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Setup

- Real Business Cycle model with financial sector *à la* Gertler and Karadi (2011)
- HH can only hold deposits and financial intermediaries invest in capital
- Financial frictions: they can divert a fraction θ of assets for private consumption
- Macro-uncertainty: TFP
- **Financial uncertainty**: random quantity of deposits is diverted from (to) the financial sector to (from) gvt bonds
  - Convenient way to shock the financial constraint
  - Idea supported by data on bank funding Chart
  - Interpretation: HH delegates portfolio decision to a noise trader.

Household

- Problem:

$$\max_{C_t, D_t, L_t} \mathbb{E}_0 \left[\sum_{t \geq 0} \beta^t \frac{(C_t - \chi L_t)^{1-\gamma}}{1-\gamma} \right]$$

s.t.

$$C_t + D_t = W_t L_t + R_{t-1} D_{t-1} + T_t$$

- Euler Equation

$$\mathbb{E}_t[\Lambda_{t,t+1}] R_t = 1$$

$$\Lambda_{t,t+1} = \beta \left(\frac{C_{t+1} - \chi L_{t+1}}{C_t - \chi L_t} \right)^{-\gamma}$$

- Labor supply

$$W_t = -\frac{\partial U_t / \partial L_t}{\partial U_t / \partial C_t} (= \chi)$$

Intermediaries

$$V_t = \max_{N_{t+1}, K_{t+1}} \mathbb{E}_t \left\{ \Lambda_{t,t+1} [(1-\sigma) N_{t+1} + \sigma V_{t+1}] \right\}$$

s.t.

$$Q_t K_{t+1} = N_t + \widetilde{D}_t \qquad \text{Flow of Funds}$$

$$N_{t+1} = R^k_{t+1} Q_t K_{t+1} - R_t \widetilde{D}_t \qquad \text{Net worth LOM}$$

$$V_t \geq \theta Q_t K_{t+1} \qquad \text{Financial constraint}$$

$\widetilde{D}_t = e^{\sigma^\nu_t \nu_t} D_t$, $\nu_t \overset{iid}{\sim} \mathcal{N}(0,1)$ IRF and σ^ν_t is **financial uncertainty**:

$$\sigma^\nu_t = (1-\rho_\sigma)\overline{\sigma^\nu} + \rho_\sigma \sigma^\nu_{t-1} + \varepsilon^\sigma_t$$

Optimality condition:

$$\mathbb{E}_t[\Lambda_{t,t+1}(1 - \sigma + \sigma \underbrace{\psi_{t+1}}_{\frac{\partial V_{t+1}}{\partial N_{t+1}}})(R^k_{t+1} - R_t)] = \mu_t \theta (= 0).$$

Production and Government

- Final good producer

$$\max_{L_t} e^{Z_t} K_t^{\alpha} L_t^{1-\alpha} - W_t L_t - Q_t I_t$$

$$\text{s.t. } W_t L_t \leq \zeta Q_t K_t$$

$$Z_t = (1-\rho_z)\bar{z} + \rho_z Z_{t-1} + \sigma^z \varepsilon_t^z$$

- K_t producers

$$\max_{I_t} Q_t I_t - \Phi\left(\frac{I_t}{K_t}\right) K_t$$

$$Q_t = \Phi'\left(\frac{I_t}{K_t}\right)$$

- Capital LOM

$$K_{t+1} = (1-\delta)K_t + I_t$$

- Government budget constraint

$$B_t + T_t = R_{t-1} B_{t-1}$$

Equilibrium

- Labor mkt equilibrium: fraction f of firms is assumed constrained

$$\chi = f\frac{\zeta Q_t K_t}{L_t} + (1-f)(1-\alpha)e^{Z_t}K_t^{\alpha}L_t^{-\alpha}$$

- Financial Markets equilibrium:

$$N_{t+1} = \sigma(R_{t+1}^k Q_t K_{t+1} - R_t \widetilde{D}_t) + \omega Q_t K_{t+1}$$
$$B_t = D_t - \widetilde{D}_t$$

- Goods mkt equilibrium

$$C_t + \Phi\left(\frac{I_t}{K_t}\right)K_t = Y_t$$

Calibration

Variable	Parameter	Value	Source
Discount factor	β	0.995	Standard
Risk aversion	γ	1	Standard
Capital share	α	0.33	Standard
Depreciation	δ	0.025	Standard
Mean of TFP	$\bar{z}$	0	Standard
Persistence of TFP	ρ_z	0.95	Standard
S.d. of TFP innovations	σ_z	0.01	Standard
Surviving bankers	σ	0.972	GK (2011)
Transfer to new bankers	ω	0.002	GK (2011)
Fraction of divertible assets	θ	0.381	GK (2011)
Disutility of labor	χ	0.4352	Internally calibrated
Fraction of constrained firms	f	0.4233	Internally calibrated
Fraction of pledgeable assets	ζ	0.0933	Internally calibrated
Mean of σ_t^ν (s.d. of ν_t)	$\overline{\sigma^\nu}$	0.0179	Internally calibrated
Persistence of σ_t^ν	ρ_σ	0.9	Internally calibrated

Solution Algorithm Calibration Details

Conditional IRF from Global Solution

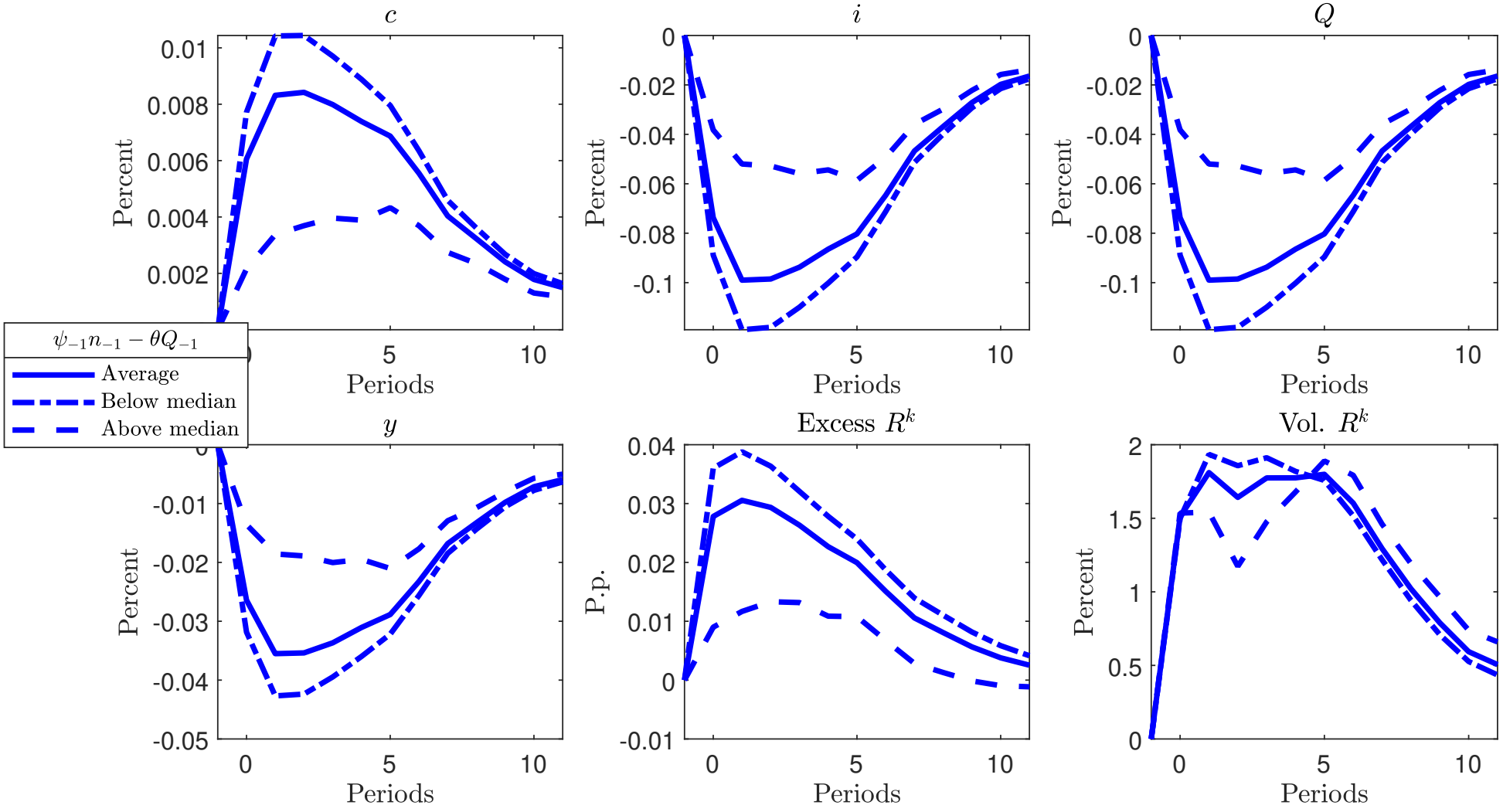

Figure: IRF to a 2× shock to σ_t^ν, conditional on the initial state of the economy.

Empirical vs theoretical IRF

Table: Ratio between IRF conditional on "distressed" and "not distressed" financial sector for selected variables.

	i		y		$\mathbb{E}[R^k] - R$	
h	Model	Data	Model	Data	Model	Data
1	1.97	1.71	1.98	1.19	2.11	1.77
2	1.58	1.43	1.59	1.32	1.68	2.17
3	1.46	1.36	1.47	1.26	1.71	3.98
4	1.43	1.66	1.44	1.6	2.07	4.37

Conclusion

- New evidence that uncertainty shocks $\times$ financial sector capitalization can explain excess returns countercyclicality
- Model with financial intermediaries and two separate sources of uncertainty, one macroeconomic and one financial
- Model successfully generates the contingency of responses
- Avenues for future research:
  - Quantitatively match model to data
  - Introduce nominal frictions
  - Micro-found noise traders
  - Policy response.

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Literature Review

- **On excess return cyclicality**
  - Fama and French (1989), Ferson and Harvey (1991), Lettau and Ludvigson (2001), Lettau and Ludvigson (2009), Backus et al. (2010), Lustig and Verdelhan (2012), Nagel and Xu (2023) for stocks;
  - Gilchrist and Zakrajšek (2012) for corporate bonds;
  - Cochrane and Piazzesi (2005) and Ludvigson and Ng (2009) for U.S. gvt bonds;
  - Lettau et al. (2014) for commodities, sovereign bonds and currencies.
- **On uncertainty shocks**
  - Bloom (2009), Basu and Bundick (2017), Bloom et al. (2018), Fernandez-Villaverde et al. (2011), Fernandez-Villaverde et al. (2015), Born and Pfeifer (2014) without credit constraints;
  - Christiano et al. (2014), Cesa-Bianchi and Fernandez-Corugedo (2018), Gilchrist et al. (2014), Fernandez-Villaverde and Guerrón-Quintana (2020), Chatterjee et al. (2020), Fang and Liu (2021) with credit constraints.
- **On intermediary-based asset pricing**
  - He and Krishnamurthy (2013), Brunnermeier and Sannikov (2014), He et al. (2017) in finance;
  - Bernanke et al. (1999) Gertler and Karadi (2011), Gertler et al. (2020) in macro.

Bloom (2009)

- Construct volatility index by splicing VXO with volatility of SP500 index
- Run a VAR with

$$\mathbf{y}_t = [R_t, vola_t, ffr_t, cpi_t, wages_t, empl_t, hours_t, ip_t]$$

- Shock identification: ordering and Choleski decomposition
- $vola_t$ shock contemporaneously affects the stock market, then prices and finally quantitites.

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Jurado Ludvigson Ng (2015) and Ludvigson Ma Ng (2021)

- 132 macro variables divided in 8 groups: income and production, labor mkt, housing, consumption and investment, money and credit, interest and exchange rates, prices;
- 147 financial variables: dividends, prices, d-p ratios, common factors and returns on a large number of portfolios;
- Estimate a FAVAR model for all the variables
- compute $h-$step ahead forecast and forecast errors;
- Uncertainty at horizon h = st. dev. of forecast errors at horizon h
- Shock identification through VAR $[U_{Mt}, ip_t, U_{Ft}]$ and "shock-based restrictions" (event constraints).

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Data and Variables I

- "Equity-invcap" variable on COMPUSTAT for all the firms in the "Financial" sector according to the GICS classification
- **Excess Returns**
  - SP500 returns from CRSP
  - Excess returns computed as the monthly 10yr return on stock minus the monthly 10yr risk free rate taken from Kenneth French website
  - GZ spread and EBP from Gilchrist and Zakrajšek (2012)
  - Term spread = 10 years-3 months Treasury rate at constant maturity. Source: FRED
  - Mortgage spread = 30 years fixed rate mortgage - 10 years Treasury rate. Source: FRED
- **Balance sheet items**
  - Credit to nonfinancial businesses by financial institutions (depository institutions and others). Source: BIS statistics. Credit includes both loans and debt securities
  - "Domestic Financial Sectors; Treasury Securities; Asset, Level", Source: FRED
  - "Domestic Financial Sector; Total Financial Asset, Level" and "Total Financial Liabilities" to compute net worth. Soruce: FRED
  - "All Sectors; Total Mortgages; Asset, Level". Source: FRED

Strategy Specification

Data and Variables II

- Controls for excess returns: $ip_t, UNC_t, \log Q_t, \pi_t$ (FRED), common factors (Kenneth French website)
- Controls for macro aggregates: $ip_t, FFR_t, 10Y - 3m, \pi_t, \log Q_t$ (FRED)
- **Debt Instruments** (see Gorton et al. (2012))
  - "Money Market Funds; Total Financial Assets"
  - "Domestic Financial Sectors; Checkable Deposits and Currency"
  - "Domestic Financial Sectors; Federal Funds and Security Repurchase Agreements"
  - "Private Depository Institutions; Total Time and Savings Deposits"
  - "Finance Companies; Commercial Paper"
  - "Private Depository Institutions; Bankers' Acceptances"
  - "Domestic Financial Sectors; Total Miscellaneous Liabilities"
- Bank funding data from Bowman, Scotti, and Vojtech (2020), who use Call Report and FR-Y9C.

Specification IRFs

Uncertainty indexes

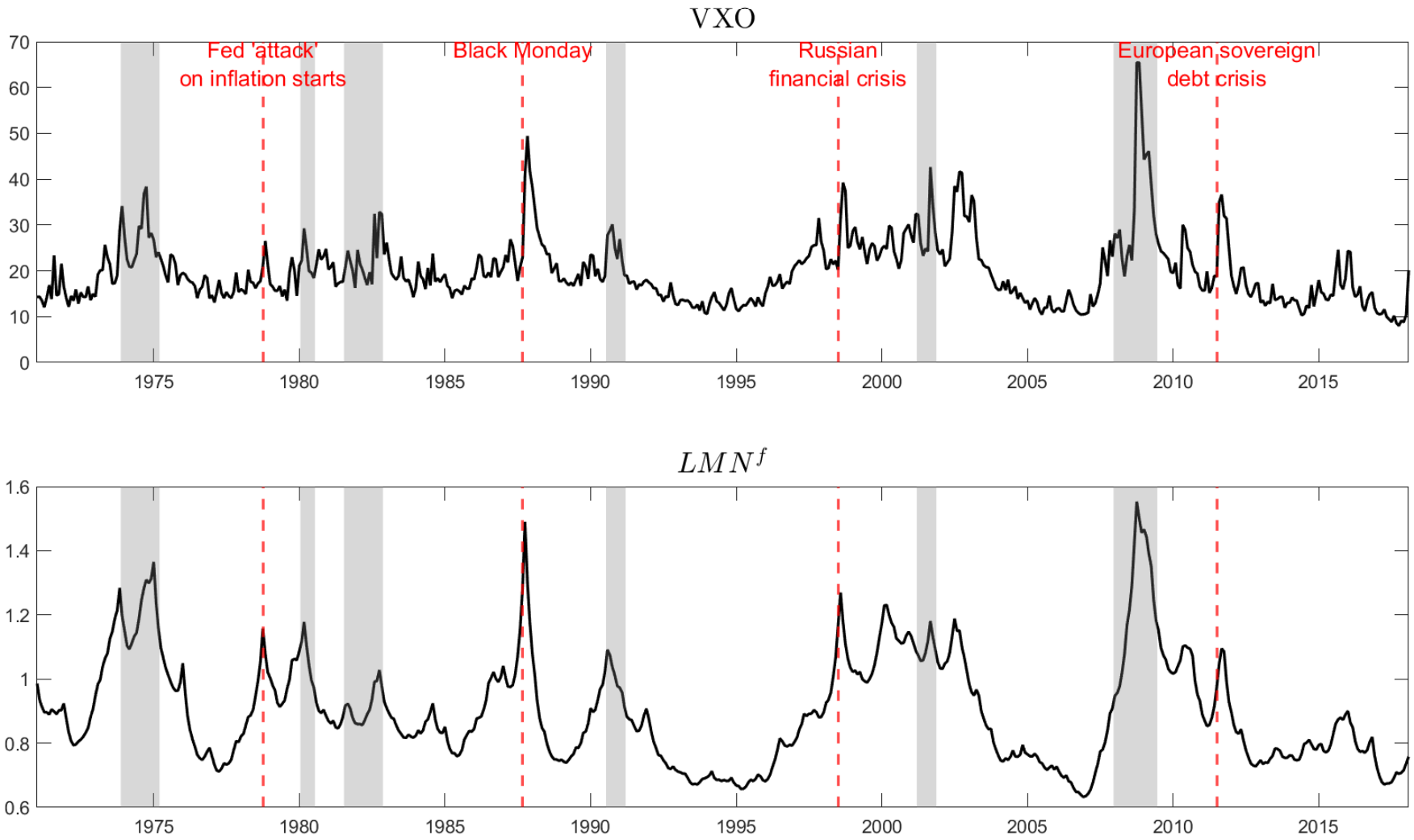

Figure: VXO and U_F^{LMN} indexes with significant events in financial markets history. Back

Excess Returns: shock to U_F^{LMN}

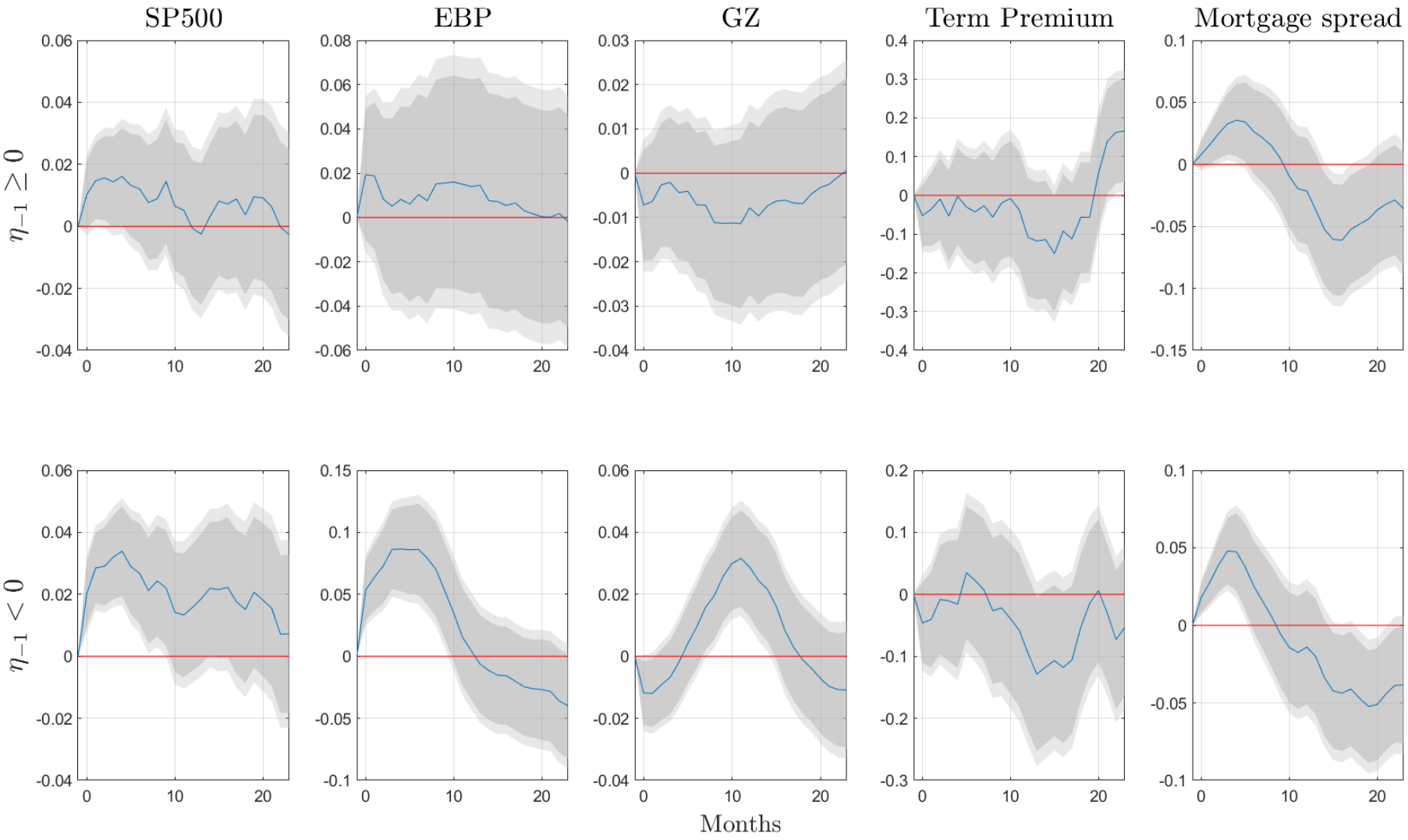

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Back

Real Variables: shock to U_F^{LMN}

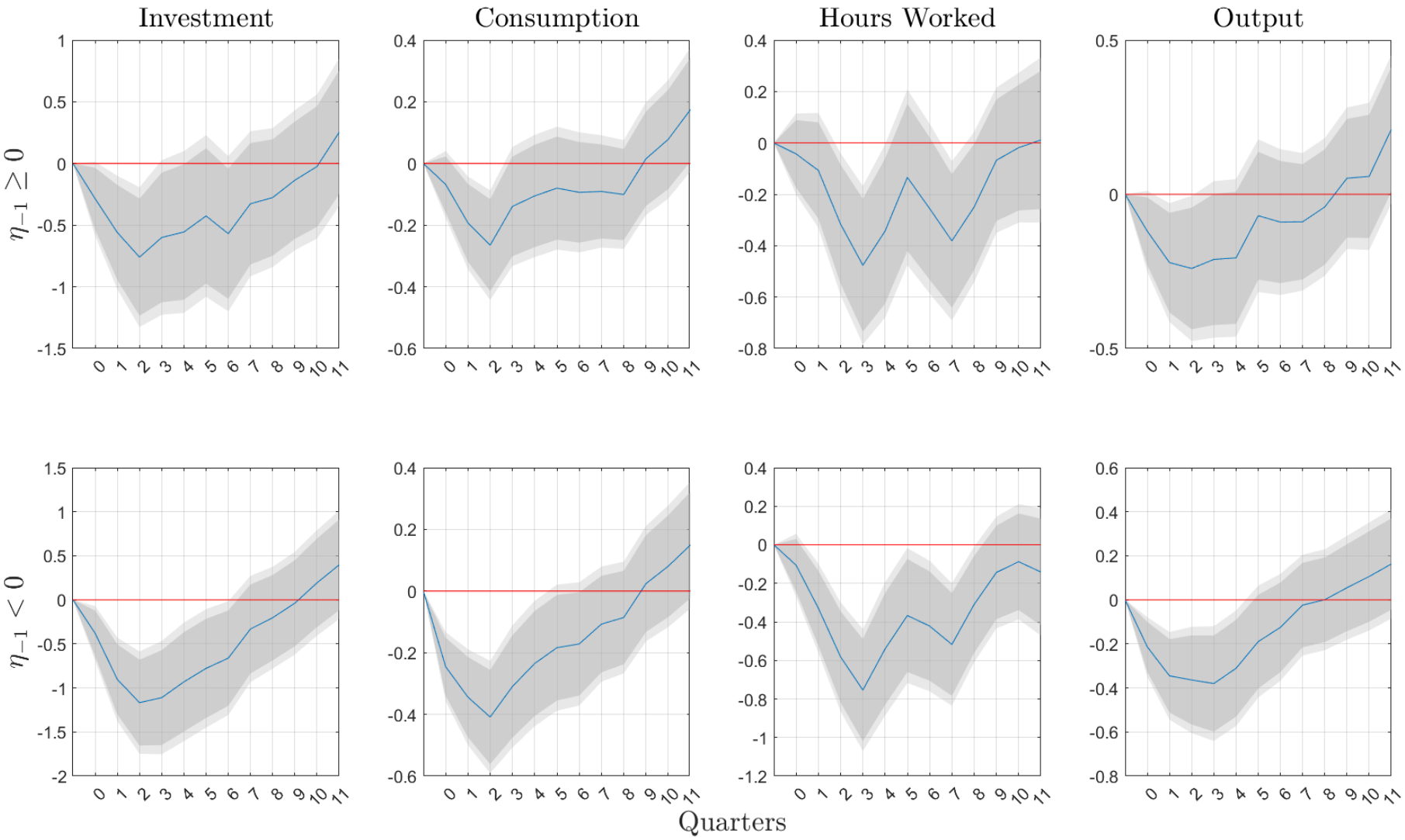

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Back

Financial Variables: shock to U_F^{LMN}

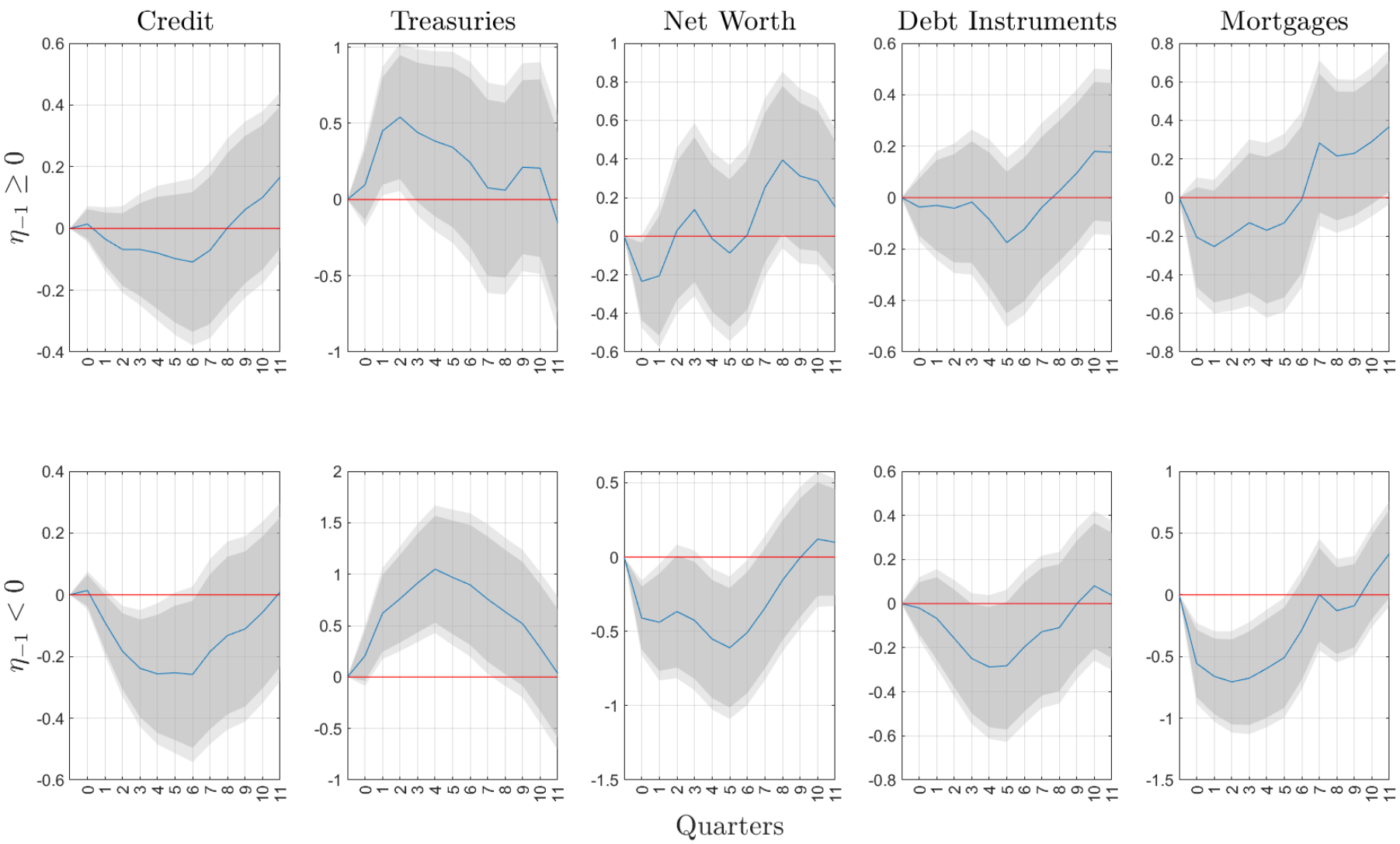

Figure: Dark grey: 90% confidence bands. Light grey: 95% confidence bands. Y-axis in percentage points. Back

Solution Strategy I

- Normalize quantity variables by K and denote $x = \frac{X}{K}$
- States: $S = (\log p, Z)$ with $p_{t-1} = R_{t-1} D_{t-1}$
- Eq. Functions $\Theta = (c, \ell, u, i, y, R, \psi, Q, n, p')$, where $u = c - \chi\ell$
- (ℓ, R, ψ) pin down (c, u, i, y, Q, n, p'). Static Equations

Solution Strategy II

(ℓ, R, ψ) come from the solution of the system:

$$\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')((R^k)'-R)\right]=\theta\mu \quad (1)$$

$$\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}\right]R=1 \quad (2)$$

$$\psi=\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\frac{n'}{n}\right] \quad (3)$$

$$\mu(\psi n-\theta Q)\geq 0 \quad (4)$$

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Solution Strategy III

Rewrite Equation (3) and condition (4) as

$$\mu = \max\left\{1 - \beta \frac{u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\right] Rn}{\theta Q}, 0\right\} \tag{5}$$

$$\psi = \max\left\{\beta u^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(u')^{-\gamma}(1-\sigma+\sigma\psi')\right] R, \frac{\theta Q}{n'}\right\} \tag{6}$$

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Algorithm

- Start with small value of θ: solution shouldn't be too far from unconstrained (RBC) solution
- Initial guess ℓ_0: RBC solution; R_0 derived from (2) and $\psi_0 = \max\left\{\frac{\theta Q_0}{n_0'}, 1\right\}$
- Given (ℓ_n, R_n, ψ_n) update ℓ by solving for (1) and (5);
- Given $(\ell_{n+1}, R_n, \psi_n)$, update R using (2) and ψ using (6);

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Model Static Equations

- $Q = \frac{\chi\ell - (1-f)(1-\alpha)e^{Z}\ell^{1-\alpha}}{f\zeta}$
- $y = e^{Z}\ell^{1-\alpha}$
- $i = (\Phi')^{-1}(Q) = a_1^{-\frac{1}{\xi}} Q^{\frac{1}{\xi}}$
- $c = y - \Phi(i)$
- $n'(1-\delta+i) = \sigma\left\{[\alpha y + (1-\delta)Q] - p\right\} + \omega Q$
- $p' = R(Q - n')$

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Calibration Details

Table: Target statistics for parameter calibration. Shaded parameters are determined jointly.

Parameter	Target statistic	Model	Data
f	$\frac{Var(\hat{Q}_t)}{Var(\hat{\ell}_t)}$	2	2.23
ζ	$\frac{Q_{ss}}{\ell_{ss}}$	2.52	2.43
χ	$\frac{K}{L}$	1.34	1.4
$\overline{\sigma^{\nu}}$	$\frac{s.d.}{avg}$ g.r. of total bank funding	0.0179	0.0179
ρ_{σ}	IRF persistence for R^{exc}_{SP500}	0.9023	0.988

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Bank funding

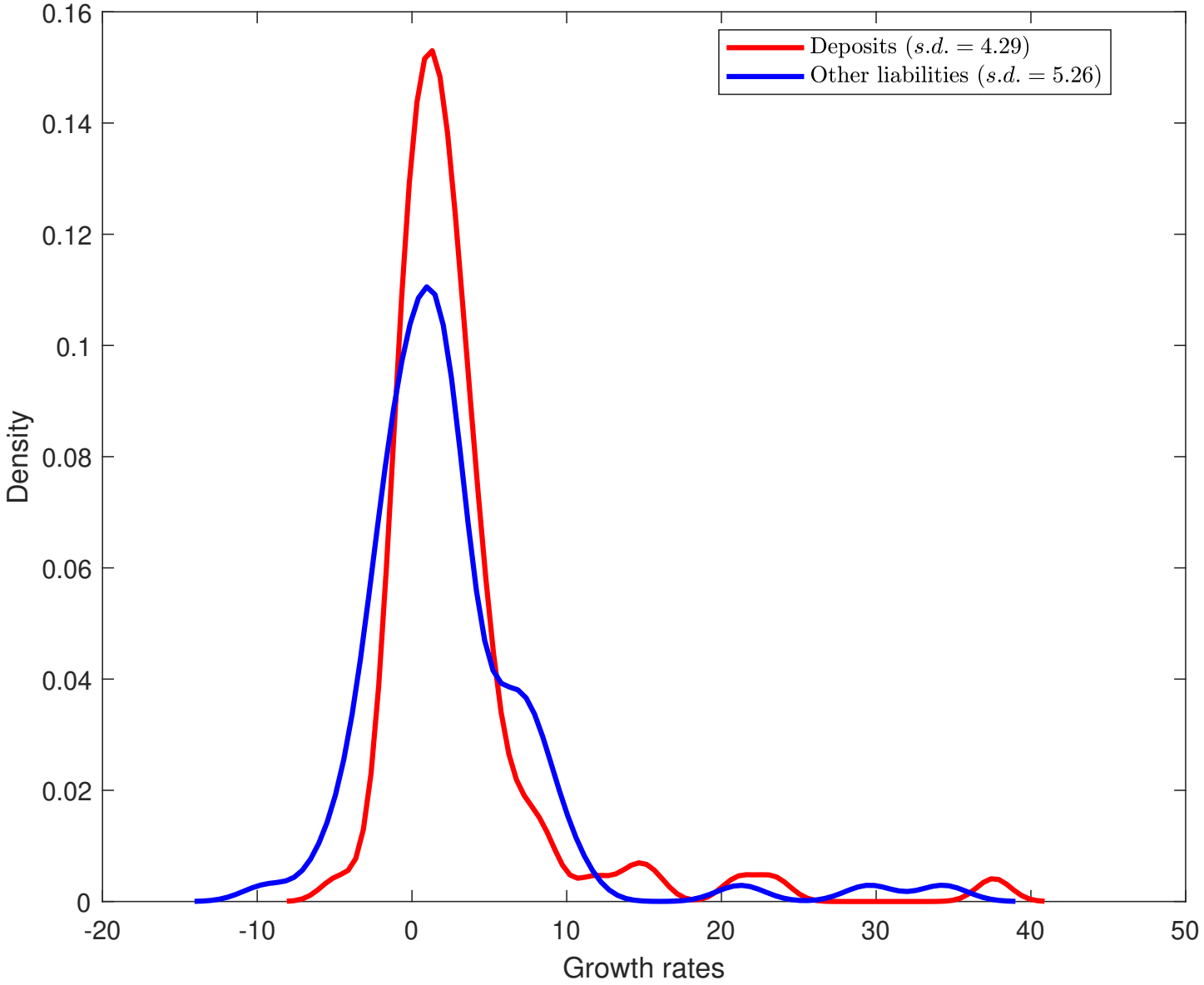

Figure: Kernel density of growth rates of bank funding by type. Data source: Bowman, Scotti, and Vojtech (2020), FR-Y9C, Call Report. Back

Dynamics of a shock ν_t

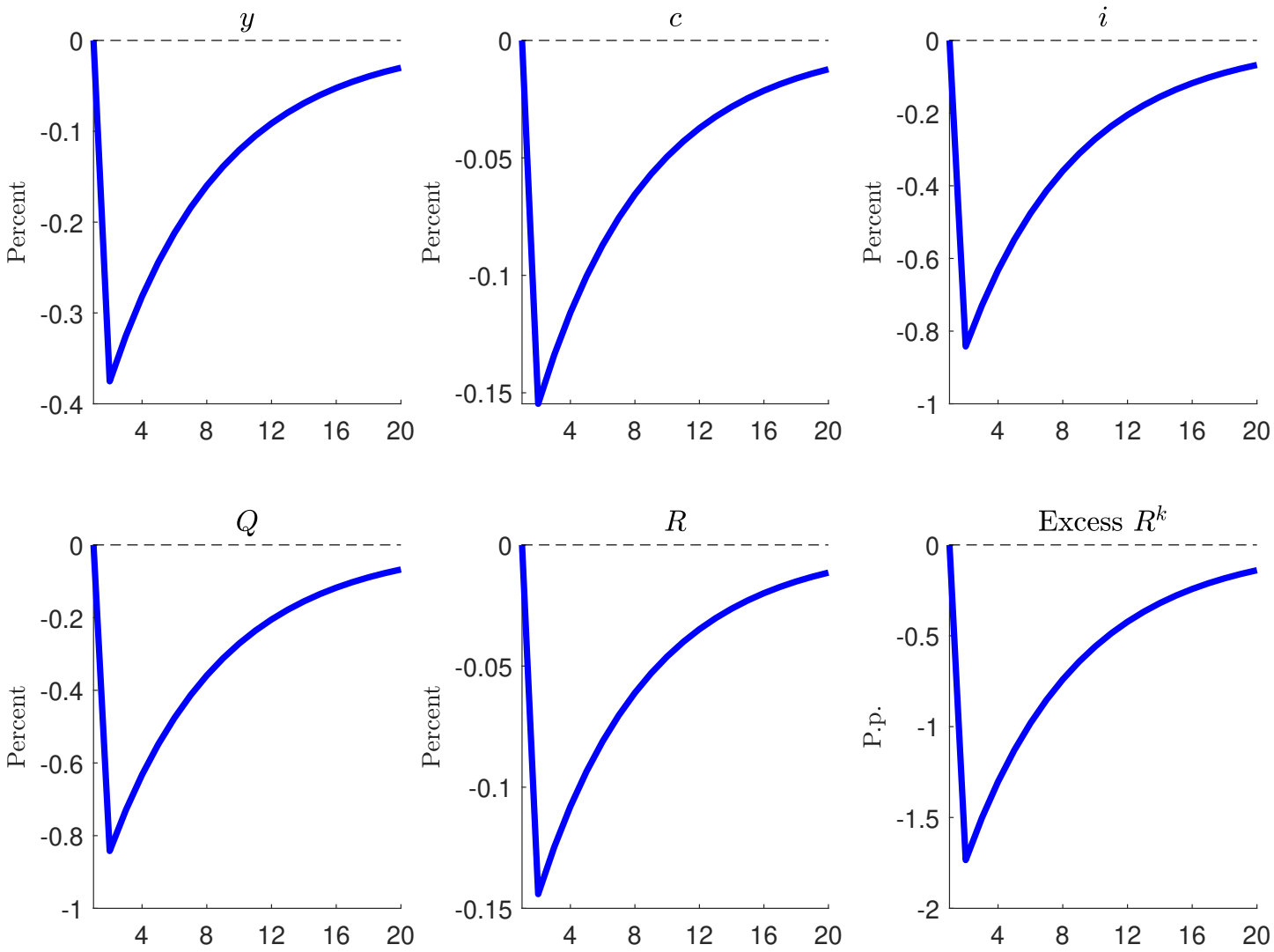

Figure: IRF to 1 s.d. shock to ν_t. Back